Introduction to R

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Topics

- What is R?
- R pros and cons
- What working with R looks like
- Use R with the RStudio Development Environment
  - Get help / documentation
  - Work with contributed packages
  - Read and store data
  - Explore / transform / aggregate data
  - Create diagrams
  - Descriptive statistics
  - Statistical tests & regression models
- Integrated documents

Showcase

- Radiation dose to the heart in 3D conformal tangential breast cancer radiotherapy
- Dashboard structure comparison
- Report generation

What is R?

What is R? www.r-project.org

- Statistical environment
  - Free
  - For interactive use
  - Command-driven
- *Base R* = interactive console
  - vs. Integrated development environment (IDE)
  - vs. Contributed extension packages
- (Scripting programming language)

Pros: Why use R?

- **It works** (Windows, MacOS, Linux, Cloud)
- Widely used, active development & community
  - MSE: Students increasingly already have skills in R
- Free & open source
  - No cost / no licensing issues → use anywhere
  - Supports autonomy
    - Typically, the limiting factor is your skill, not R
- Powerful extension mechanism: Packages
  - Fast moving innovation covering many use cases
    - Example: `https://pharmaverse.org/`

Cons: What's not that great about R

- First steps easy but hard to master
  - Use it or lose it `r4stats.com/articles/why-r-is-hard-to-learn/`
- Organic growth → Inconsistent naming scheme
  - `read.table()`, `seq_int()`, `TukeyHSD()`, `trimws()`
- Built-in documentation very terse, technical
- There's more than one way to do it
  - Packages → inconsistency & confusion
    - Instability
- Handling very large data requires skill

Switching from SAS – key differences

- R is case sensitive
- No restriction to working with just 1 data set
  - Always be explicit about which data you are using
- Foundational data structure is a vector (single variable)
- Different philosophy
  - SAS: Few functions, many options, much output
  - R: Many functions, few options, little output
- Data does not write itself to disk automatically
  - Need to explicitly save data
- Base R: No metadata on variable labels, lengths, formats
- Dummy coding categorical variables: ref = 1st level

Switching from SAS – Online resources

- https://www.dst.dk/Site/Dst/SingleFiles/GetArchiveFile.aspx?fi=35925166405 56&fo=0&ext=direktion
- https://www.youtube.com/watch?v=8_5n_yHufrA
- https://www.sasconverter.com/resources
- https://www.appsilon.com/post/gsk-r-adoption-journey
- https://www.youtube.com/watch?v=xDrt6txplek (GSK Journey to adopt R)
- https://www.youtube.com/watch?v=BlJNILSoZlM (Roche Journey to adopt R)
- https://bayer-group.github.io/sas2r/
- https://github.com/brendanjodowd/cheatsheets/blob/master/sas-r.pdf
- https://r-sassy.org/
- https://pharmasug.org/proceedings/2024/HT/PharmaSUG-2024-HT-201.pdf
- https://pharmaverse.org/

Should you learn R?

- Results matter
  - Tools matter only insofar as they enable results
- Can you afford to **continuously** invest time?
- Do you have a specific project / task?

Documentation: Systematic

- Integrated help system with executable examples
- Official introduction and FAQs `www.r-project.org`
- Books
  - `https://link.springer.com/book/10.1007/978-3-662-71786-8`
  - `https://link.springer.com/book/10.1007/978-3-662-63075-4`
  - `https://dwoll.de/r/rinfo.php`
- Online courses
  - `www.edx.org/learn/r-programming/harvard-university-statistics-and-r`
  - `https://cs50.harvard.edu/r/`
- Cheat sheets
  - `https://posit.co/resources/cheatsheets/`

Learn R: Unsystematic documentation

- Google `"R"`
  - Many tutorials, blog posts, ...
- Claude / Copilot / ChatGPT
  - Integrate into RStudio

Beware

Quality issues – wrong / outdated info?

Using R in RStudio

Run R

- Get R from `cloud.r-project.org/`
- Get a free integrated development environment (IDE)
  - **RStudio** ← `posit.co/download/rstudio-desktop/`
    Supported: Windows, MacOS, Linux, Cloud
  - Up and coming? Positron `https://positron.posit.co/`
  - Used to programming? → Zed `https://zed.dev/`

Start

- R session: Working directory
- Use the integrated help system
- Objects
  - Create, show & remove objects
  - Classes: vector, matrix & data frame
  - Types: character, numerical, logical, factor, date / time
- Arithmetic
- Logic: Operators → associativity, ( )
- Functions: Arguments & return values

https://github.com/dwoll/vebik_r/

Start – Vocabulary

- `getwd()`, `setwd()`
- `?⟨function⟩`, `??⟨word⟩`, `help()`, `help.start()`, `example()`
- `?Reserved`, `<-`, `=`
- `print()`
- `ls()`, `rm()`
- `?Arithmetic`
- `round(⟨number⟩, digits=⟨digits⟩)`
- `exp()`
- `TRUE`, `FALSE`, `!`, `==`, `!=`, `<`, `>`, `<=`, `>=`, `|`, `&`

Hands on – run R within RStudio

- Do some arithmetic
- Store an integer each in objects a, b, add them
- Is 1 the same as `"1"`? Why?
- What arguments can be supplied to function `mean`?
- Run the examples for `var()`
  - What is going on – how is this all related to `var()`?

Frequent sources of confusion & frustration

- Mis-spelled objects, functions, arguments (case-sensitive!)
  - Look at suggestions & use tab-completion
- Whitespace is often meaningless – but not always
- Handling of missing values – NA (not available)
- Handling of categorical variables (factors)
- Execution order (*associativity*) → use ( ) liberally

Using R packages

R packages

- Extend base R functionality for a specific purpose
  - Specialized statistical analysis
  - Tools for specialized data
  - More convenient solutions than base R
- Bundle new functions, data sets and documentation
- Contributed by independent developers
- Have dependency management

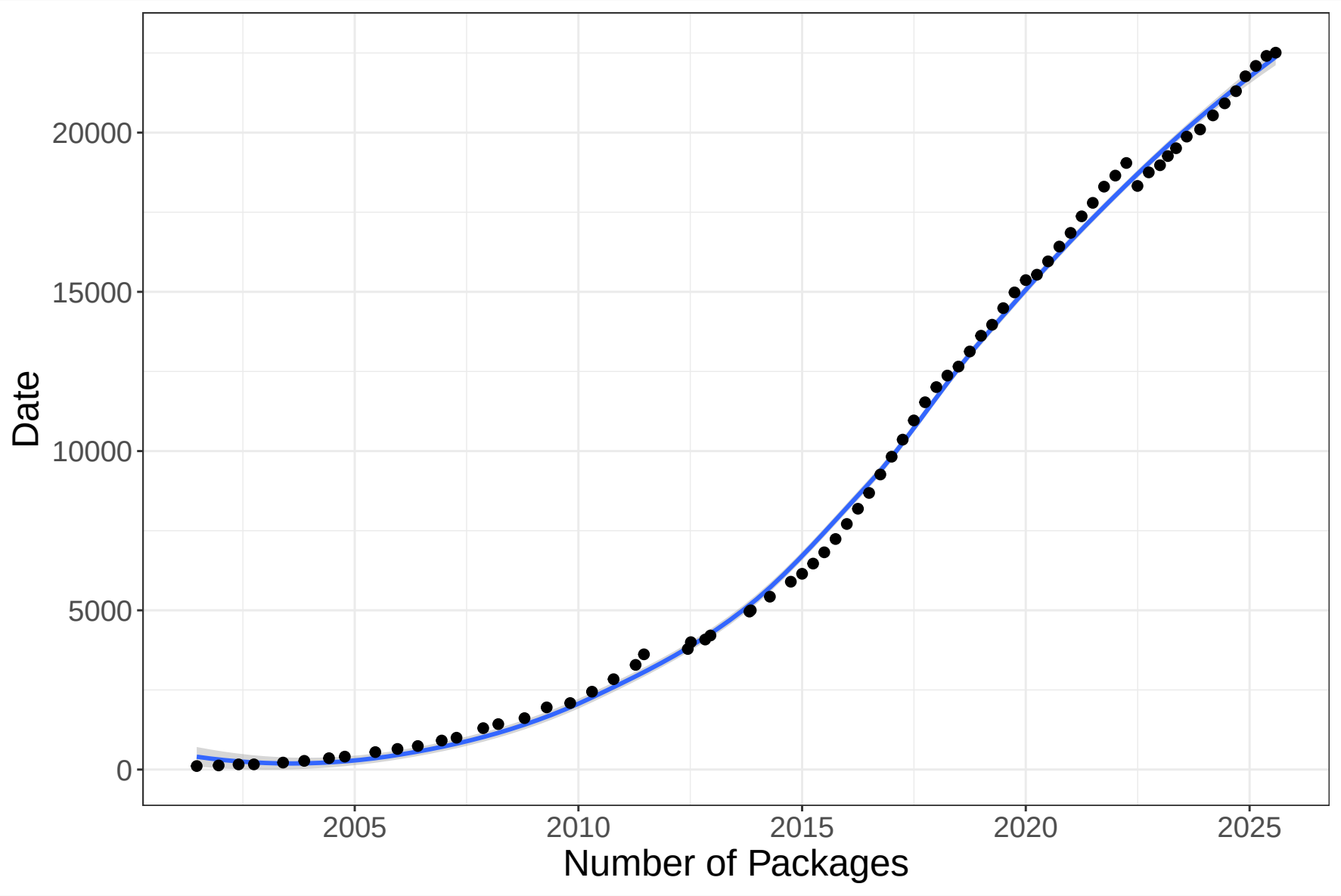
Using packages

1. Install – once for every major R version
2. Load – each session
3. Use like any base R functionality

R package repositories

- CRAN
  - Curated official package repository network
  - Some quality assurance for submitted packages
    - Can be installed, run, are documented, not malicious
    - **But:** No guarantee for working correctly
- GitHub
  - Not curated, no QA, bleeding edge

Growth of CRAN package repository



Using packages – Vocabulary

- `install.packages("⟨package name⟩")`
- `installed.packages(), .libPaths()`
- `update.packages()`
- `library(⟨package name⟩)`
- `sessionInfo()`
- `help(package="⟨package name⟩")`
- `vignette(package="⟨package name⟩")`
- `vignette("⟨topic⟩")`
- `citation("⟨package name⟩")`

Hands on

- Load package `dplyr`
- What help topics are documented in package `ggplot2`?
- Look at the `dpylr` vignette *Introduction to dplyr*
- Run the examples for `filter()` from package `dplyr`
- Check if there are updates for your installed packages
  - But don't install them

Read & store data

- Local files & URL
- Plain text file: Comma-separated, tab-delimited, ...
- R format file
- SAS / Stata / SPSS file: package `haven`
- Excel: package `readxl`, `openxlsx`
- Database: `RSQLite`, `RPostgreSQL`, `duckdb`, ...

Read & store data – Vocabulary

- `read.table()`, `write.table()`
- `file=url()`
- `readRDS()`, `saveRDS()`
- Package `readxl`
  - `read_xlsx()`
- Package `openxlsx`
  - `write.xlsx()`
- Package `haven`
  - `read_sas()`, `read_spss()`

Read & store data – Hands on

- Read text file
  - `https://dwoll.de/dat_sossap.csv`
  - View data in RStudio data explorer
  - Save the data set to an R file
  - Save the data set as a tab-separated ASCII file
- Read Excel file
  - `https://dwoll.de/dat_sossap.xlsx`
- Find a SAS or SPSS file and read it in

Explore data – Vocabulary

- `dim()`, `nrow()`, `ncol()`
- `head()`, `tail()`, `names()`, `View()`
- `str()`
- `[["⟨VarName⟩"]]`, `$`

Explore data – Hands on

- How large is the PASSOS data set?
- What variables of what kind are there?
- Take a look at the first / last 10 observations
- Extract variable patID

Transform data

- Select subsets
  - Cases
  - Variables
  - Remove duplicates
- Change & create new variables
  - Recode variables
  - Cut continuous variables into categories
  - Calculate new based on old variables
  - Generate sequences
  - Simulate using random numbers
- Sort cases & variables

Transform data – Vocabulary

- Package `dplyr`
  - `filter()`
  - `select()`, `everything()`
  - `mutate()`, `rename()`, `if_else()`
  - `arrange()`, `desc()`
- `unique()`, `na.omit()`
- `factor()`
- `scale()`, `cut()`, `as.Date()`
- `:`, `seq()`, `rep()`
- `sample()`, `runif()`, `rnorm()`

Transform data – Hands on

- Check and transform the PASSOS data set
  - Sort according to tumor side and age
  - Subset: Only women age ≥ 60 with left-sided tumor
  - Re-order variables: *observed* comes 1st, *metric* 2nd
  - Build age-groups (10 years)
  - Do a median split for BMI
  - Center `observed`
  - Make all date variables actual dates
  - Make `Seite` a factor

Diagrams

- Books
  - *R Graphics Cookbook* `www.r-graphics.org`
- Base graphics
- Package `ggplot2` ←

Diagrams – Vocabulary

- ggplot2
  - ggplot(), aes()
  - group=, linetype=, shape=, color=, fill=
  - geom_point(), geom_line(), geom_bar()
  - geom_histogram(), geom_density(), geom_boxplot()
  - geom_smooth()
  - position=position_jitter(), position_dodge()
  - scale_x_continuous(), scale_y_continuous()
  - labs()
  - facet_grid(), facet_wrap()
  - theme()
- ggsave()

Diagrams – Hands on (PASSOS data set)

- Look at `metric == "DMEAN"` and `structure == "1_HEART"`
- Histogram of `observed`
- Boxplot by `Seite`
- Scatter plot of `observed` by `BMI`
  - Diagram title, axis labels
  - Choose axis limits and grid spacing
  - Add linear fit
  - Point color and shape by `Seite`
  - Change to `B/W` theme
- Separate panels for `Mainz / Ulm`

Summarise data

- Descriptive statistics
- Group wise operations
- Frequency tables

Summarise data – Vocabulary

- Package `dplyr`
  - `group_by()`, `ungroup()`
  - `summarise()`, `n()`
- `sum()`, `min()`, `max()`, `range()`, `diff()`, `quantile()`
- `mean()`, `median()`, `sd()`, `var()`, `IQR()`
- `with()`
- `cov()`, `cor()`
- `xtabs()`, `as.data.frame()`
- `proportions()`, `addmargins()`

Summarise data – Hands on (PASSOS data set)

- Look at `metric == "DMEAN"` and `structure == "1_HEART"`
- Group by `center` and `Seite`
  - Number of observations, mean and SD of `observed`
- Frequency table `center` and `Seite`
  - Absolute & conditional relative frequencies (by row)
- Correlation matrix of `Alter`, `BMI`, `observed`

Statistical models

- Linear regression (OLS)
- GLM: Logistic, Poisson regression
- *t*-test, ANOVA, ANCOVA
- Survival: Kaplan-Meier, Cox regression

Statistical models – Vocabulary

- `lm()`, `summary()`
- `coef()`, `confint()`
- `glm()`
- `t.test()`, `alternative=`, `wilcox.test()`, `anova()`, `aov()`
- `chisq.test()`, `fisher.test()`, `binom.test()`
- Package `survival`
  - `survfit()`, `survdiff()`, `coxph()`

Statistical models – Hands on (PASSOS data set)

- Look at `metric == "DMEAN"` and `structure == "1_HEART"`
- Is `observed` higher for left-sided tumors?
- Two-way ANOVA of `observed` by `Seite` and `haveLAG`
- Linear regression of `observed` on `Alter` and `BMI`
- Do some regression diagnostics on the fit
- Logistic regression model for Median-split BMI group by `Alter`

Data containers

- Vectors
  - Recycling
- Factors
- Data frames
- Lists

Data containers – Vocabulary

- `c()`
- `[]`, `length()`
- `%in%`
- `factor()`
- `nlevels()`, `levels()`, `droplevels()`, `interaction()`
- `data.frame()`, `list()`

Specific data types

- Missing values
- Character strings
- Date & time

Specific data types – Vocabulary

- Missing values
  - `NA`, `is.na()`, `anyNA()`, `na.omit()`
  - `na.rm=TRUE`, `use="pairwise"`, `use="complete"`
- Character strings
  - `nchar()`, `trimws()`, `tolower()`, `toupper()`
  - `paste()`, `paste0()`, `sprintf()`
  - `grepl()`, `gsub()`, `glob2rx()`
  - Package `stringr`
- Date & time
  - `Sys.date()`, `as.Date()`, `strptime()`
  - Package `lubridate`

Advanced data transformations

- Split
- Combine
- Merge
- Transform between wide ↔ long format

Advanced data transformations – Vocabulary

- `split()`, `lapply()`
- `dplyr`
  - `bind_rows()`, `bind_cols()`
  - `left_join()`
- `reshape()`

Create documents & share results with Quarto

- https://quarto.org/docs/get-started/
- https://quarto.org/docs/guide/
- ...see .qmd file

R PROGRAMMING – BASICS

- Books
  - *Advanced R*: `https://adv-r.hadley.nz/`
  - *R packages*: `https://r-pkgs.org/`
- No loops, but: `lapply()`
- Control structures
  - Conditions: `if() { … } else { … }`

R PROGRAMMING – FUNCTIONS

- Function signature
  - Formal arguments and defaults
- Function body
  - Check actual arguments
  - Handle errors
  - Return value
  - Scope
- Generic functions
- Analyze functions
- Debugging